KK-L-159

DECEMBER 6, 1956

SUPERSEDING
Int. Fed. Spec. KK-L-00191a(COM-NBS)
June 24, 1955, and
Fed. Spec. KK-L-191
June 16, 1952
(See 6.4)

FEDERAL SPECIFICATION

LEATHER, CATTLEHIDE, (INSOLE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers insole leather in the form of cut insoles made from cattlehide and intended for use in the types of footwear in which a leather insole is required.

1.2 Classification.

1.2.1 *Types.*—Insole leather, in the form of cut insoles, covered by this specification shall be furnished in the following types and tannages, as specified in the invitation for bids:

Type I. — Combination chrome-vegetable tanned.
Type II.—Straight vegetable.

1.2.2 *Classes.*—The insole leather, in the form of cut insoles, shall be furnished in the following classes, as specified in the invitation for bids:

Class 1.—Men's insoles.
Class 2.—Women's insoles.

1.2.3 *Thickness.* — The insole leather, in the form of cut insoles, shall be of the thickness specified in the invitation for bid (see 3.4 and 6.1).

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standard, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

KK-L-165 — Leather, Cattlehide, Vegetable-Tanned; and Soles, Leather.
KK-L-311—Leather, Methods of Sampling and Testing.
LLL-B-631—Boxes; Fiber, Corrugated, (For Domestic Shipment).
LLL-B-636 — Boxes; Fiber, Solid (For Domestic Shipment).
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes; Wood, Nailed and Lock Corner.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

JAN-P-108—Boxes, Fiberboard, Corrugated and Solid (Military Type).

FSC Class 8330

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MIL–L–10547 — Liners, Case, Waterproof.

MIL–B–10377 — Box, Wood, Cleated, Veneers, Paper Overlaid.

Military Standard:

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by the contractor in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Type I.*—The leather shall be made from cattlehides and shall be tanned by a process containing chrome followed by vegetable retannage.

3.1.2 *Type II.*—The leather shall be made from cattlehides and shall be tanned by a process commercially known as vegetable tannage.

3.2 Mold resistance. — Unless otherwise specified, the leather shall be treated with paranitrophenol.

3.3 Finish.—The leather shall be finished full grain, degrained or buffed with the flesh side smooth and free from loose fibers.

3.4 Thickness.—The leather shall be of the thickness specified in the invitation for bids. In general, class 1 insoles will be required in the range of 5½ to 7½ irons and class 2 insoles in the range of 3 to 5 irons.

3.4.1 *Thickness tolerance.*—The thickness shall be not less than that specified, but a plus tolerance of 1 iron will be permitted.

3.5 Defects. — The leather shall be free from "not acceptable" defects as listed in table I.

3.6 Cutting areas.—Insoles from any area in a hide meeting all requirements of this specification are considered to be satisfactory.

TABLE I.—*Class of Defects**

Class of defects	Acceptable	Not acceptable
Brands	Light brand on heel	Brand on shank or forepart or heavy brand in heel area.
Wrinkles	Slight fat wrinkles not hingy	Heavy fat wrinkles.
Scratches	Slight scratches; healed scratches	Deep scratches.
Grain damage	Slight grain damage; well healed tick scars	Grain damage which will not buff out. Extensive cracked grain in shank or forepart. Open tick scars.
Grub holes	Healed grub scars	Open grub holes or healed grub scars in channel area.
Hardness		Hard and bony insoles.
Slaughter cuts	Slight cuts or scores on flesh, if perpendicular to channel area, providing that it does not affect the serviceability of the shoe	Deep cuts. Slight cuts parallel to and within the channel area.
Skiving	Limited amounts of skiving or leveling	Excess skiving.
Fleshiness	Slightly fleshy	Fleshy channeling area. Excess flesh or nap.
Flexibility		Flexibility not uniform. Excessive stiffness.
Pattern damage	Insoles filling the sole pattern	Any pattern damage.
Fiber characteristics	Medium-tight	Hard and very tight. Loose and spongy.

* See Federal Specification KK–L–165 for figures illustrating defects.

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3.7 Chemical requirements.

3.7.1 *Type I.*—The leather shall conform to the requirements listed in table II.

TABLE II.—*Chemical requirements, moisture-free basis*

Characteristics	Minimum	Maximum
Degree of tannage	35.0	--
Water solubles, percent	--	26.0
Grease (chloroform extract), percent	3.0	10.0
Acidity, pH	3.0	4.0
Paranitrophenol, percent	0.18	0.40
Chrome, Cr_2O_3, percent	2.0	--

3.7.2 *Type II.*—The leather shall conform to the requirements listed in table III.

TABLE III.—*Chemical requirements, moisture-free basis*

Characteristics	Minimum	Maximum
Degree of tannage	60.0	--
Grease (chloroform extract), percent	3.0	10.0
Water soluble material, percent	--	35.0
Insoluble ash, percent	--	1.0
Acidity, pH	3.0	4.0
Paranitrophenol, percent	0.18	0.40

3.8 Physical requirements. — The leather shall conform to the following physical requirements:

3.8.1 *Area stability.*—The leather shall not shrink more than 10 percent in area when tested in accordance with 4.3.3.1.

3.8.2 *Cracking.* — The leather shall not crack when bent grain out over a 3-inch mandrel (see 4.3.4).

3.8.3 *Piping.*—The leather shall not pipe when bent grain in over a 3-inch mandrel (see 4.3.4).

3.8.4 *Shrinkage temperature (type I only).* —The shrinkage temperature of the leather shall be not less than 90°C. (see 4.3.4).

3.9 Workmanship.—Workmanship shall be such as to produce leather in the form of cut insoles complying with the requirements of this specification.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.

4.1.1 *Formation of inspection lots.* — Lots of each type, class and thickness shall be formed as described in 3.1 of section 6 of Federal Specification KK–L–311.

4.1.2 *Sampling for inspection.* — Samples shall be taken for inspection as described in 3.2 of section 6 of Federal Specification KK–L–311.

4.1.3 *Sampling for physical and chemical tests.* — Samples for physical and chemical tests shall be taken as described in 3.3 of section 6 of Federal Specification KK–L–311 for fabricated leather, except that pieces for physical and chemical tests shall be taken from each test unit (cut insole) in the sample for test, not closer than ½ inch to any edge.

4.2 Inspection.

4.2.1 Each unit in the sample for inspection shall be inspected for type, class, defects, and thickness.

4.2.2 *Thickness.* — The thickness shall be determined as described in 4.1.3 (sole leather) of method 1011 of Federal Specification KK–L–311.

4.3 Tests.

4.3.1 *Specimens for physical tests.*—Specimens for physical tests shall be taken from

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the test units as required in the applicable method of test.

4.3.2 *Specimens for chemical tests.*—Specimens for chemical tests shall be taken from the composite sample prepared as described in method 6002 of Federal Specification KK–L–311 as required in the applicable method of test.

4.3.3 *Methods of test.*

4.3.3.1 *Area stability.*—To determine conformance with 3.8.1, a specimen 2 by 2 inches shall be cut from the body of the insole, immersed in sufficient distilled water (68° to 78°F.) to cover the specimen for 2 hours at room temperature (68° to 78°F.), then removed, blotted on a towel and air-dried at room temperature for 24 hours. The area shall then be measured and the change in area calculated.

4.3.4 *Other tests.* — Other physical and chemical tests shall be made in accordance with the following methods described in Federal Specification KK–L–311.

Test	Method No.
Cracking	4021
Piping	4111
Shrinkage temperature	7011
Thickness	1011
Organic Solvent (Grease) using chloroform as solvent	6321
Water soluble material	6341
Insoluble ash	6411
Nitrogen	6441
Paranitrophenol	6711
Formaldehyde	6111
Hide substance	6461
Combined tannin	6631
Degree of tannage	6641
Acidity (pH)	6621
Chromic oxide	6511 or 6515
Aluminum oxide	6531

4.4 Acceptance of lot.—A lot of leather shall be considered as meeting the requirements of this specification if the conditions described in 5.1 of section 6 of Federal Specification KK–L–311 are fulfilled.

5. PREPARATION FOR DELIVERY

5.1 Nonmilitary.

5.1.1 *Packaging.*—Unless otherwise specified, commercial packages are acceptable.

5.1.2 *Packing.*—Unless otherwise specified, the subject commodity shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

5.1.3 *Marking.*

5.1.3.1 *Issue packages.*—Unless otherwise specified, issue packages shall be marked with the name, type, class, and thickness of the material, the quantity contained therein, and the names of the contractor and manufacturer.

5.1.3.2 *Shipping containers.*—Unless otherwise specified, shipping containers shall be marked with the name, type, class, and thickness of the material as defined by the contract or order under which the shipment is made, the quantity of each type, class, and thickness contained therein, the name of the contractor, the number of the contract or order, the name and address of the consignee, and the gross weight.

5.2 Military.

5.2.1 *Packaging.*

5.2.1.1 *Level A—Military package.*—Unless otherwise specified, the insoles shall be packaged in accordance with the manufacturer's standard practice. When specified in the contract or order, ten (10) insoles of one size only, shall be packaged together by wrapping in kraft paper.

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5.2.1.2 *Level B — Limited package.* — The insoles shall be packaged as specified in 5.2.1.1.

5.2.1.3 *Level C—Minimum Military Package.*—The insoles shall be packaged as specified in 5.2.1.1.

5.2.2 *Packing.*

5.2.2.1 *Level A — Military pack.* — The insoles, packaged as specified in 5.2.1.1 shall be packed in boxes constructed, closed and strapped in accordance with any one of the following specifications and appendices thereto: style A or B of Federal Specifications PPP–B–601; PPP–B–621; or compliance symbol V3c or V3s of Military Specification JAN–P–108. Gross weights of wood boxes shall not exceed 100 pounds; that of fiberboard boxes shall not exceed 70 pounds. Unless otherwise specified, each box shall be provided with an unsealed case liner in conformance with type I, grade B or C, class 2 of Military Specification MIL–L–10547.

5.2.2.2 *Level B—Limited Military pack.*— The insoles packaged as specified in 5.2.1.2, shall be packed in boxes conforming with any one of the following Federal or Military Specifications: Style A or B (domestic type) of PPP–B–601; style 2 or 4 of PPP–B–621, MIL–B–10377, domestic type; LLL–B–631 or LLL–B–636. When fiberboard boxes are used, the fiberboard shall meet the special requirements of either Federal Specification LLL–B–631 or LLL–B–636, as applicable. Gross weights of wood-cleated or wood boxes shall not exceed 100 pounds; that of fiberboard boxes shall not exceed 65 pounds.

5.2.2.3 *Level C—Minimum Military pack.* —The insoles, packaged as specified in 5.2.1.3 shall be packed in boxes which are acceptable by common or other carriers for safe transportation to point of destination specified in shipping instructions at the lowest transportation rate for such supplies.

5.3 Marking.—Interior packages and shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Ordering data. — Purchasers should specify the type, class, and thickness required and should exercise any desired options offered herein (see 1.2, 3.2, 5.1, 5.2.1.1, 5.2.2.1). Military agencies should also specify whether the insoles shall be packaged and packed for Level A—Military, Level B—Limited Military, or Level C — Minimum Military described as follows:

(1) *Level A—Military.*—Applies to that material to be transported by ocean-going or coastwise vessel and handled, issues and stored in the military supply system.

(2) *Level B—Limited Military.*—Applies to that material for storage and reshipment to a domestic receiving activity not involving sea transportation.

(3) *Level C — Minimum Military.* — Applies to that material required for immediate use without storage or reshipment by the initial domestic receiving activity.

6.2 Transportation description.—The transportation description applicable to this item is:

Sole leather cut stock.
Carload minimum weight 30,000 pounds, subject to Rule 34, Uniform Freight Classification.

Sole leather.
Truckload minimum weight 24,000 pounds.

6.3 Federal Standard No. 102, Preservation, Packaging, and Packing Levels, should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.

6.4 Supersession.—This specification supersedes Military Specification MIL–L–11636B, Leather, Cattlehide, Insole.

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MILITARY INTERESTS:
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